1. a) i. let, Sajib is using 'n' bit.

ATQ, $\frac{488 \times 100}{2^{12}-1} = \frac{122 \times 100}{2^n - 1}$

$\Rightarrow 2^n - 1 = \frac{122 \times 100 (2^{12}-1)}{488 \times 100} \Rightarrow 2^n = 1024.75$

$\therefore \boxed{n = 10}$

ii) ADC reading $= \frac{(2^8-1) \times 3.37}{5} = 171.87 \simeq 172$

Duty cycle $= \frac{172 \times 100}{2^8-1} = 67.45\%$

2. There are four mistakes.

i. Start → should be sent from master.

ii. 'E' should be received instead of 'e'.

iii. Before stop, NACK should be sent instead of ACK.

iv. Stop should be sent by the master.

3. Von-neuman architecture should be used.

Because,

i. Low-cost

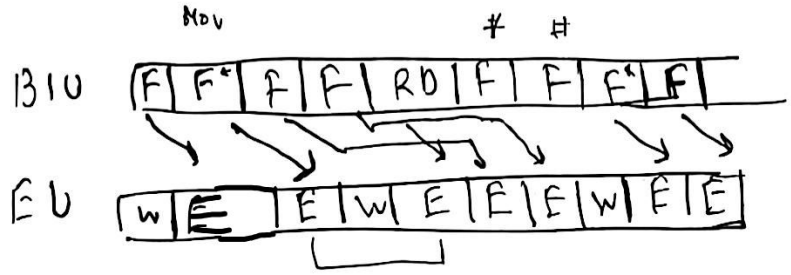
ii. Less complex

CamScanner

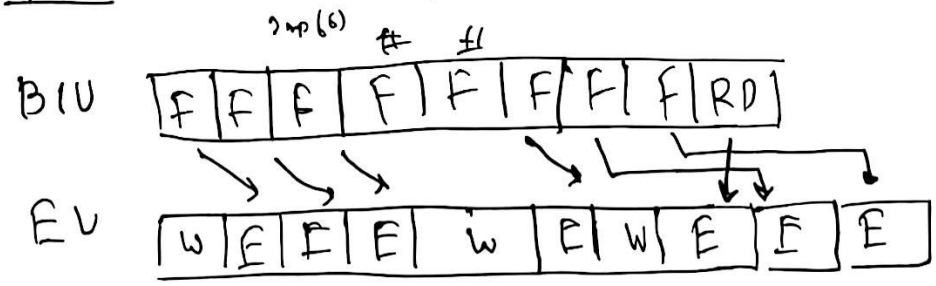
(2)(a)i 1st → Jmp(2)

2nd → Mov

4th → Jmp(7

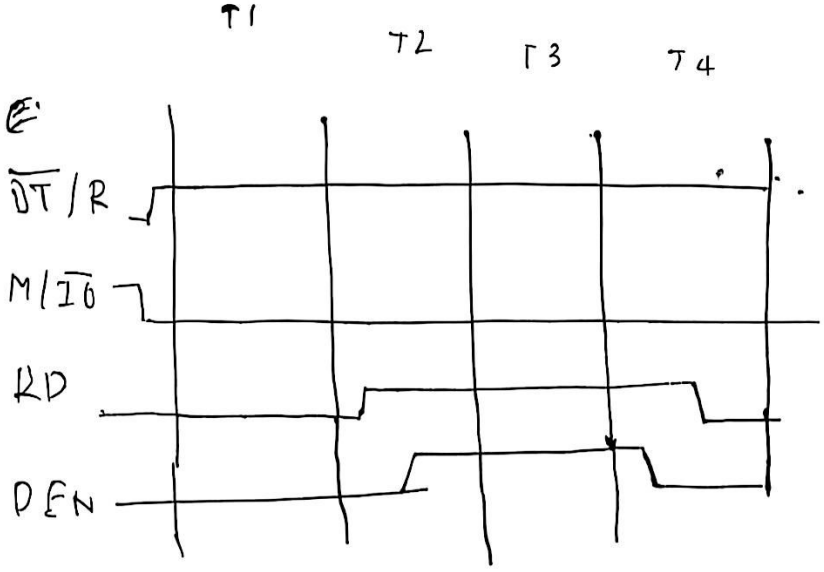


2(a)ii 3rd → Jmp(6) 5th → ↑(6), 6th → Read



(2b)

$\overline{DT}/R$, DEN, $M/\overline{IO}$, RD



Dsec Dexlansoprazolo PANSEC Pantoprazolo Pronex-MUPS Esomoprazolo Pronex Esomoprazolo

3) a)

```
0H:  SUB  EH
1H:  ADD  9H
2H:  OUT
3H:  ADD  CH
4H:  LDA  DH
5H:  SUB  BH
6H:  HLT
7H:  ADD  AH
```

3) b)

Instruction Address	Accumulator Register	B Register
Start	04H	01H
0H	04-02=02H	02H
1H	02H+29H=2BH	29H
2H	2BH	29H
3H	2BH+04H=2FH	04H
4H	3CH	04H
5H	3CH-15H=27H	15H
6H	00H	00H
7H	00H	00H

3] c) I)

$CON = E_u \bar{S}_u \bar{L}_o \bar{L}_B \; \bar{L}_I \bar{E}_I \bar{L}_A E_A \; E_p C_p \bar{L}_M CE$

CON =

Disable: 0 1 1 1 1110 0010

= 7E2

For ADD:

T4: Intruction Reg $\xrightarrow{WBUS}$ MAR

CON = 0111 1010 0000

= ~~7B0~~ 7A0

T5: RAM $\xrightarrow{WBUS}$ B Reg

CON = 0110 1110 0011

= 6E3

T6: $\bar{S}_u$ Select, Adder: $\xrightarrow{WBUS}$ Acc

CON = 1111 1100 0010 [$\bar{S}_u = 1$ for ADD]

= FC2

3) c) For LDAOUT, the T5 state will be same as the T5 state of LDA operation, which is, the content of RAM ~~goes~~ is loaded in Accumulator Register through WBUS.

In T6, it will be same as the T4 state of OUT operation, which is, the content of accumulator goes to output register using WBUS.

∴ T5, CON = 0111 1100 0011

= 7C3

T6, CON = 0101 1110 0010

= 5F2

3) d) $-3-5\times4 = 4-7-4-4-4-4-4$

Already loaded in Acc. Reg. -3

∴ Assembly Code:

```
0H: SUB AH
1H: SUB CH
2H: SUB CH
3H: SUB CH
4H: SUB CH
5H: SUB CH
6H: OUT
7H: HLT
```